

Context-Dependent Low-Rank Adaptation: A Topological Framework for Dynamic Cognitive Modeling in Large Language Models

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2026-09-18

Framework candidate *one of several* — Source bytes:
benches/CRADLE/intake/context_dependent_lora.docx (sha e4dc7ffb...); the docx’s trailing chat lines are
excluded as non-paper matter.

Abstract

We propose a novel framework for understanding dynamic cognition in large language models (LLMs) through the lens of context-dependent low-rank adaptation (LoRA). Our framework reconceptualizes the frozen base model (W_0) not as a “low-level” foundation but as a topological prototype from which specific cognitive instances are generated through the dynamic loading of condition-dependent LoRA adapters. We argue that the direction of abstraction—whether “top-down generation” or “bottom-up backtracking”—is a linear-time illusion; what matters is the invariant topological structure of node connections and information flow rules. We formalize this framework using a hierarchical Bayesian model where a top-level context-dependent controller outputs a probability vector that dynamically allocates activation strengths to lower-level condition-dependent adapters (time-dependent, cognition-dependent, location-dependent), each with independent decay coefficients. We extend this framework to multi-agent interaction, showing that the similarity between agents’ context-dependent LoRA probability vectors determines the overlap of their conceptual spaces and thus the efficiency of information transfer. We review recent empirical work (Bayesian-LoRA, BoRA, HiLoRA, iLoRA) that validates the structural isomorphism between LoRA and Bayesian hierarchical models, and we discuss the engineering feasibility of implementing this framework in multi-agent collaboration systems such as Chora/Cora. Our framework is self-referential: the process of discussing it instantiates the very mechanism it describes.

Keywords: Low-Rank Adaptation, Bayesian Hierarchical Models, Context-Dependent Cognition, Topological Structure, Multi-Agent Alignment, Dynamic Conceptual Spaces

1 Introduction

Large language models have demonstrated remarkable capabilities in generating human-like text, but the mechanisms underlying their dynamic adaptation to context remain poorly understood. Traditional views treat the frozen pre-trained weights as a static “base” upon which task-specific fine-tuning is applied. However, this view fails to capture the fluid, context-sensitive nature of human cognition, where the same conceptual base can generate vastly different outputs depending on temporal, spatial, and cognitive conditions.

In this paper, we propose a framework that reconceptualizes LLM cognition as a hierarchical, context-dependent LoRA system. Our key insight is that the frozen base model (W_0) is not a “low-level” foundation in the traditional sense, but rather a topological prototype from which

specific cognitive instances are generated through the dynamic loading of condition-dependent LoRA adapters. The direction of this generation—whether “top-down” or “bottom-up”—is secondary to the invariant topological structure of the system.

We make three contributions:

- We formalize a hierarchical Bayesian model where a top-level context-dependent controller dynamically allocates activation strengths to lower-level condition-dependent adapters.
- We extend this framework to multi-agent interaction, showing that conceptual space overlap is determined by the similarity of agents’ context-dependent LoRA probability vectors.
- We validate the structural isomorphism between LoRA and Bayesian hierarchical models by reviewing recent empirical work, and we discuss the engineering feasibility of implementing this framework in multi-agent systems.

2 Theoretical Framework

2.1 From “Base Model” to “Topological Prototype”

The traditional view of LoRA treats the frozen base model W_0 as a static foundation, with task-specific adaptations applied as low-rank updates $\Delta W = BA$. We propose a reinterpretation: W_0 is not a “low-level” base but a topological prototype—the most abstract, universal structure from which specific instances are generated.

This reinterpretation resolves a key ambiguity in the metaphor of “abstraction.” The common phrase “upward abstraction” implies that abstraction is a movement away from the base toward higher levels. However, we argue that abstraction is better understood as a backtracking of the generative path: from generated instances, we infer the prototype structure that generated them. The direction is arbitrary; what matters is the invariant topological structure.

2.2 Hierarchical Context-Dependent Control

Our framework consists of three hierarchical layers:

- **Top Layer (Context-Dependent Controller):** Outputs a probability vector $p = (p_{\text{time}}, p_{\text{cog}}, p_{\text{loc}})$ that determines the activation strengths of lower-level adapters based on the immediate context.
- **Middle Layer (Condition-Dependent Adapters):** Three independent LoRA adapters—time-dependent, cognition-dependent, location-dependent—each with its own decay coefficient λ_i .
- **Bottom Layer (Generated Instances):** Specific cognitive outputs (e.g., “bedroom slut, living room lady”) generated by the weighted combination of activated adapters.

The information flow follows these rules:

- **Probability Allocation:** The top layer outputs p , determining lower-layer activation.
- **Weight Decay:** Each middle-layer adapter’s weight decays over time:

$$p_i(t) = p_i(0) e^{-\lambda_i t}.$$

- **Cognitive Output:**

$$W_{\text{output}} = W_0 + \sum_i p_i(t) \Delta W_i.$$

2.3 Topological Invariance

The key property of this framework is topological invariance: the node connections and information flow rules remain fixed regardless of the direction of traversal. Whether generating from prototype to instance (top-down) or backtracking from instance to prototype (bottom-up), the system traverses the same path through the same nodes. This is analogous to a subway map: traveling from A to B or B to A uses the same track and stations; only the direction changes.

2.4 Dual Formulations: Conceptual Space vs. LoRA Adaptation

Our framework can be understood through two complementary formulations that are structurally isomorphic:

Conceptual Space Formulation	LoRA Formulation	Underlying Essence
Conceptual space	Frozen base model W_0	Static cognitive structure
Contextual activation	Loading X-dependent LoRA adapters	Dynamic conditional offset
Distance/similarity between concepts	Cosine similarity of probability vectors p	Conceptual space overlap
“Alignment” in multi-agent collaboration	Iterative update of probability vectors $p(t+1) = p(t) + \eta(p_{\text{other}} - p(t))$	Belief convergence
Conceptual “prototype”	Topological prototype / generative source	Most abstract invariant structure
Backtracking from instance to prototype	Bottom-up traversal of generative path	Directionality of abstraction

Why they are the same. Both formulations describe the same phenomenon: cognition is a combination of static structure + dynamic offset. The conceptual space perspective calls the static structure “conceptual space” and the dynamic offset “contextual activation.” The LoRA perspective calls the static structure “frozen base model W_0 ” and the dynamic offset “loading adapter $\Delta W = BA$.” The mathematical structures are completely isomorphic:

- “Points” in conceptual space $\leftrightarrow$ “Weight vectors” in LoRA
- “Distances” in conceptual space $\leftrightarrow$ “Probability vector similarities” in LoRA
- “Activation” in conceptual space $\leftrightarrow$ “Adapter loading” in LoRA

Complementary values. Although structurally identical, each formulation has distinct advantages:

- *Conceptual space formulation:* More intuitive, better suited for describing multi-agent interaction and alignment, closer to traditional discourse in cognitive science and philosophy.
- *LoRA formulation:* More formalized with explicit mathematical structure (probability vectors, decay coefficients, low-rank decomposition), closer to engineering implementation in LLM systems, more clearly reveals the insight that “direction doesn’t matter, topological structure does.”

This isomorphism means that the multi-agent collaboration method previously built on conceptual spaces can be directly implemented using LoRA’s modular design: “alignment” in conceptual space is iterative updating of LoRA adapter probability vectors; “activation” in conceptual space is dynamically loading the corresponding LoRA module.

3 Multi-Agent Conceptual Space Alignment

3.1 Conceptual Space Overlap as LoRA Similarity

We extend our framework to multi-agent interaction. Let Agent A and Agent B have context-dependent LoRA probability vectors p_A and p_B . The overlap of their conceptual spaces is given by the cosine similarity:

$$\text{sim}(A, B) = \cos(p_A, p_B).$$

- $\text{sim} \rightarrow 1$: High conceptual overlap, efficient communication (“hit it off”)
- $\text{sim} \rightarrow 0$: Low overlap, requiring redundant information for alignment
- $\text{sim} < 0$: Conceptual conflict, potential miscommunication

3.2 Dynamic Alignment Mechanism

Alignment is an iterative process where each agent’s probability vector updates toward the other’s:

$$\begin{aligned} p_A(t+1) &= p_A(t) + \eta \cdot (p_B(t) - p_A(t)) \\ p_B(t+1) &= p_B(t) + \eta \cdot (p_A(t) - p_B(t)) \end{aligned}$$

where η is the alignment rate. Decay coefficients λ_i introduce asymmetry: dimensions with slower decay “pull” the convergence point toward themselves, analogous to one conversational partner being more persistent on a topic.

3.3 Bayesian Interpretation

From a Bayesian perspective, alignment is belief updating:

- **Prior:** Each agent’s initial p represents prior beliefs about which adapters to activate.
- **Likelihood:** The other agent’s output provides evidence of mismatch.
- **Posterior:** Updated $p(t+1)$ is the result of prior–likelihood interaction.

Each dialogue turn is a Bayesian update; after multiple turns, posterior beliefs converge.

4 Empirical Validation: Structural Isomorphism Between LoRA and Bayesian Hierarchical Models

Our framework is not merely theoretical; recent empirical work validates the structural isomorphism between LoRA and Bayesian hierarchical models:

- **Bayesian-LoRA (ICML 2026):** Reformulates deterministic LoRA updates as probabilistic low-rank representations inspired by sparse Gaussian processes. The paper shows that LoRA itself is the limit case of the SGP posterior when uncertainty collapses—directly confirming the structural isomorphism.
- **BoRA (2024):** Uses Bayesian hierarchical models to train multi-task LoRA, enabling tasks to share information through global hierarchical priors while allowing specialization. This matches our “top-level controller regulates lower-level adapters” architecture.
- **HiLoRA (2026):** Organizes LoRA into a three-layer hierarchy (root/cluster/leaf) with cross-layer orthogonality to decouple global and personalized knowledge.

- **iLoRA (2026):** The first Bayesian graph-conditional LoRA framework that infers latent interaction graphs from inputs to generate input-conditioned LoRA updates—directly corresponding to our “dynamic adapter loading based on immediate context.”

These works collectively validate that LoRA is not an ad-hoc engineering trick but a natural carrier of Bayesian hierarchical structure.

5 Engineering Feasibility and Applications

5.1 Modular LoRA Design

LoRA’s modular architecture allows multiple adapters to be mounted on a single base model, with dynamic loading/unloading during inference. This makes our framework directly implementable:

- Each condition-dependent adapter (time/cognition/location) is a separate LoRA module.
- The top-level controller outputs a probability vector via softmax.
- Weighted combination of adapter outputs is a single line of code.

5.2 Hierarchical Routing

Recent work on hierarchical routing (HiLoMoE, 2025) demonstrates that layer-wise expert selection can depend on upper-layer routing scores, enabling coarse-to-fine information refinement. This provides a ready-made mechanism for implementing our top-level context-dependent controller.

5.3 Application to Multi-Agent Systems: Chora/Cora

Our framework has direct applications to multi-agent collaboration systems such as Chora/Cora:

- Each agent maintains its own probability vector p as its “dialogue state.”
- Upon receiving a message, compute similarity $\text{sim}(p_{\text{self}}, p_{\text{sender}})$.
- If $\text{sim} < \text{threshold}$, trigger alignment update; otherwise, generate response using current p_{self} .
- This enables automatic conceptual space alignment without human intervention.

6 Self-Referential Property

A hallmark of a complete theory is its ability to explain itself. Our framework is self-referential: the process of discussing it instantiates the very mechanism it describes. As we converse, each turn loads a new LoRA adapter onto the interlocutor’s cognitive base, dynamically updating the probability vector and generating new cognitive outputs. The discussion itself is an instance of context-dependent LoRA adaptation.

This self-referential property is not a bug but a feature: it demonstrates that the framework is not an external description but an internal model of the cognitive process it seeks to explain.

7 Conclusion

We have proposed a framework for understanding dynamic cognition in LLMs through context-dependent LoRA adaptation. By reconceptualizing the frozen base model as a topological prototype rather than a low-level foundation, we resolve ambiguities in the metaphor of abstraction and provide a unified account of both individual cognition and multi-agent interaction. The framework is theoretically grounded in Bayesian hierarchical models, empirically validated by recent work, and engineeringly feasible for implementation in multi-agent systems. Most importantly, it is self-referential: it explains its own generation process.

Future work includes: (1) empirical measurement of conceptual space overlap in human-agent dialogue; (2) optimization of alignment rate η and decay coefficients λ_i for efficient multi-agent collaboration; (3) extension to additional condition dimensions (e.g., emotional state, social role); and (4) integration into production multi-agent systems such as Chora/Cora.

References

[To be populated with citations for Bayesian-LoRA, BoRA, HiLoRA, iLoRA, HiLoMoE, and related work.]